Efficient-Market Hypothesis

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*The **efficient-market hypothesis (EMH)** is a hypothesis in financial economics that states that asset prices reflect all available information. A direct implication is that it is impossible to "beat the market" consistently on a risk-adjusted basis since market prices should only react to new information. The idea is closely associated with Eugene Fama, in part due to his influential 1970 review of the theoretical and empirical research.*

Overview

Because the EMH is formulated in terms of risk adjustment, it only makes testable predictions when coupled with a particular model of risk. As a result, research in financial economics since at least the 1990s has focused on market anomalies — that is, deviations from specific models of risk.

The idea that financial market returns are difficult to predict goes back to Bachelier, Mandelbrot, and Samuelson. The EMH provides the basic logic for modern risk-based theories of asset prices, and frameworks such as consumption-based asset pricing and intermediary asset pricing can be thought of as the combination of a model of risk with the EMH.

Suppose that a piece of information about the value of a stock (say, about a future merger) is widely available to investors. If the price of the stock does not already reflect that information, then investors can trade on it, thereby moving the price until the information is no longer useful for trading. In the competitive limit, market prices reflect all available information and prices can only move in response to news, creating a very close link between EMH and the random walk hypothesis.

Three Forms of Market Efficiency

In his influential 1970 review paper, Eugene Fama categorized empirical tests of efficiency into three forms, referring to the information set used in the statement "prices reflect all available information."

Form	Information Set	Description
Weak	Historical prices	Stock prices already reflect all information contained in past trading data, including price history
Semi-Strong	All public information	Prices adjust rapidly to all publicly available information, including financial statements, news, a
Strong	All information (public & private)	Prices reflect all information, including insider information. Even investors with privileged acces

Historical Development

Early Origins (1900–1950s)

The concept of market efficiency had been anticipated at the beginning of the twentieth century in the dissertation submitted by Louis Bachelier (1900) to the Sorbonne for his PhD in mathematics. In his opening paragraph, Bachelier recognized that "past, present and even discounted future events are reflected in market price, but often show no apparent relation to price changes." Bachelier's work drew on the random walk model of Jules Regnault, and his thesis is now considered pioneering in the field of financial mathematics.

Research by Alfred Cowles in the 1930s and 1940s suggested that professional investors were in general unable to outperform the market. During the 1930s–1950s, empirical studies focused on time-series properties and found that US stock prices followed a random walk model in the short term.

In 1945, F. A. Hayek argued in "The Use of Knowledge in Society" that markets were the most effective way of aggregating information dispersed among individuals. Given the ability to profit from private information, self-interested traders are motivated to acquire and act on it, contributing to more efficient market prices.

The 1960s and Fama's Formalization

The efficient-market hypothesis emerged as a prominent theory in the mid-1960s. Paul Samuelson had begun to circulate Bachelier's work among economists. In 1964, Bachelier's dissertation and related empirical studies were published in an anthology edited by Paul Cootner. In 1965, Eugene Fama published his dissertation arguing for the random walk hypothesis, and Samuelson published a proof showing that if the market is efficient, prices will exhibit random-walk behavior.

In 1970, Fama published a landmark review of both the theory and the evidence. The paper extended and refined the theory and included the canonical definitions for weak, semi-strong, and strong forms of financial market efficiency. The efficient markets theory was not widely popular until the 1960s, when the advent of computers made it possible to compare calculations and prices of hundreds of stocks quickly and effortlessly.

Theoretical Foundation

How efficient markets are linked to the random walk theory can be described through the fundamental theorem of asset pricing. This theorem provides mathematical predictions regarding the price of a stock, assuming that there is no arbitrage — that is, no risk-free way to trade profitably. If arbitrage is impossible, the theorem predicts that the price of a stock is the discounted value of its future price and dividend.

If we assume the stochastic discount factor is constant and no dividend is being paid, taking logs and assuming the Jensen's inequality term is negligible implies that the log of stock prices follows a random walk (with a drift). Although the concept of an efficient market is similar to the assumption that stock prices follow a martingale, the EMH does not always assume that stocks follow a martingale.

Criticisms and Challenges

Behavioral Finance

Behavioral economists attribute imperfections in financial markets to a combination of cognitive biases such as overconfidence, overreaction, representative bias, and information bias. These have been researched by psychologists Daniel Kahneman, Amos Tversky, and Paul Slovic, and economist Richard Thaler. Empirical evidence has been mixed, but has generally not supported strong forms of the EMH.

Defenders of EMH maintain that behavioral finance actually strengthens the case for EMH in that it highlights biases in individuals and committees, not competitive markets. Manifestations of hyperbolic discounting in competitive market prices would invite arbitrage, quickly eliminating any vestige of individual biases. Similarly, diversification, derivative securities, and hedging strategies assuage if not eliminate potential mispricings from loss aversion.

Notable Critics

Warren Buffett

Argued against EMH in his 1984 presentation "The Superinvestors of Graham-and-Doddsville". Points to the preponderance of value investors among top-performing money managers as evidence against the 'luck' explanation. Nonetheless, he recommends index funds for most investors.

George Soros

Has disputed the efficient-market hypothesis both empirically and theoretically, arguing that market prices reflect biased views of participants.

Paul Samuelson

Argued the stock market is "micro efficient" but not "macro efficient": the EMH is much better suited for individual stocks than for the aggregate stock market as a whole.

Charlie Munger

Stated the EMH is "obviously roughly correct" but that extreme commitment to it is "bonkers", as the theory's originators were seduced by an intellectually consistent theory that did not properly tie to reality.

Philip Pilkington

Argued in The Reformation in Economics that the EMH is actually a tautology masquerading as a theory — the banal claim that the average investor will not beat the market average.

Peter Lynch

Argued the EMH is contradictory to the random walk hypothesis: if prices are rational and based on all available data, fluctuations are not random; but if prices follow a random walk, they are not rational.

Market Anomalies

Empirical analyses have found several problems with the EMH. Early examples include the observation that small neglected stocks and value stocks (high book-to-market ratios) tended to achieve abnormally high returns relative to what could be explained by the CAPM. Tests of portfolio efficiency by Gibbons, Ross and Shanken (1989) led to rejections of the CAPM, prompting academics to move toward risk factor models such as the Fama-French 3 Factor Model.

Additional documented anomalies include: the performance of stock markets correlating with sunshine in the city where the main exchange is located; closed-end funds trading at substantial discounts or premia to net asset value; and low P/E stocks systematically generating higher returns (Dreman & Berry, 1995).

The 2008 Financial Crisis

The 2008 financial crisis led to renewed scrutiny and criticism of the hypothesis. Market strategist Jeremy Grantham said the EMH was responsible for the crisis, claiming that belief in the hypothesis caused financial leaders to have a "chronic underestimation of the dangers of asset bubbles breaking." Former Federal Reserve chairman Paul Volcker stated: "It should be clear that among the causes of the recent financial crisis was an unjustified faith in rational expectations, market efficiencies, and the techniques of modern finance."

Richard Posner backed away from the hypothesis, accusing Chicago School colleagues of being "asleep at the switch" and suggesting that "the movement to deregulate the financial industry went too far by exaggerating the resilience — the self-healing powers — of laissez-faire capitalism." Eugene Fama, however, maintained the hypothesis held up well, noting that stock prices declined prior to the recession exactly as efficient markets would predict.

Modern Developments

Artificial Intelligence and Market Efficiency

Tshilidzi Marwala surmised that artificial intelligence influences the applicability of the EMH: the greater the number of AI-based market participants, the more efficient markets become. This is linked to the idea that machine learning can eliminate human cognitive biases at scale, pushing markets closer to the strong form of the EMH.

Legal Applications

The theory of efficient markets has been practically applied in Securities Class Action Litigation. Efficient market theory, in conjunction with fraud-on-the-market theory, has been used to justify and calculate damages. In Halliburton v. Erica P. John Fund (U.S. Supreme Court, No. 13-317), the use of efficient market theory in supporting securities class action litigation was affirmed.

Samuelson's Dictum

Nobel Prize-winning economist Paul Samuelson argued that the stock market is "micro efficient" but not "macro efficient." Research published in 2005 strongly supported this dictum, acknowledging that while individual stocks may be priced efficiently relative to each other, the entire market may undergo sentiment-driven swings. This distinction remains an active area of academic inquiry.

Key Concepts

Arbitrage

The simultaneous purchase and sale of an asset in different markets to profit from price discrepancies. The impossibility of riskless arbitrage is a foundational assumption of the EMH.

Random Walk Hypothesis

The theory that stock market prices evolve according to a random walk and thus cannot be predicted. Closely linked to the EMH but not identical to it.

Joint Hypothesis Problem

Any test of market efficiency faces this problem: it is impossible to test EMH without simultaneously testing an asset pricing model. An anomaly may indicate market inefficiency or a flawed model of risk.

Stochastic Discount Factor

A factor used in asset pricing that reflects the time value of money and risk preferences of investors. Central to the mathematical derivation of the EMH from no-arbitrage conditions.

Market Anomalies

Deviations from predictions of specific risk models. Since the 1990s, financial economics research has focused on anomalies as the primary way to study market efficiency.

Fraud-on-the-Market Theory

A legal theory that relies on the EMH: in an efficient market, publicly available misstatements are incorporated into the stock price, affecting all investors who trade at the market price.

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